71. A train travels at a certain average speed for a distance of 63 km and then travels a distance of 72 km at an average speed of 6 Km/hr more then its original speed. If it takes 3 hours to complete the total journey, what is the original speed of the train in km/hr ?

(a) 24 (b) 33
(c) 42 (d) 66

Ans. (c) Let the original speed of the train = x km/hr

According to the questions $t_1 + t_2 = \frac{63}{x} + \frac{72}{x+6} \Rightarrow 3 = \frac{63(x+6)+72x}{x(x+6)}$

$x(x+6) = 21(x+6) + 24x \Rightarrow x^2 + 6x = 21x + 126 + 24x$

$\Rightarrow x^2 + 6x - 45x - 126 = 0 \Rightarrow x^2 - 39x - 126 = 0$

$\Rightarrow x^2 - 42x + 3x - 126 = 0 \Rightarrow x(x-42) + 3(x-42) = 0$

$(x-42)(x+3) = 0$

$\therefore x = 42$ and $x = -3$ not possible

Hence, the original speed of the train 42 km/hr

72. A thief running at 8km/hr. is chased by a policeman whose speed is 10km/hr. If the thief is 100m ahead of the policeman, then the time required for the policeman to catch the thief will be

(a) 2 min (b) 3 min
(c) 4 min (d) 6 min

Ans. (b) Required time $= \frac{100}{(10-8)} \times \frac{1000}{60}$

$= \frac{100 \times 60}{2 \times 1000}$

= 3 min

ENGLISH LANGUAGE/GRAMMAR & COMPREHENSION

Direction for the following 8 (eight), items – The following eight items (items 73 to 80) are based on three passages in English to test the comprehension of English language and therefore these items do not have Hindi version.

Read each passage the answer the items that follows.

English Passage–1

Seven-year-old Jim came home from the park without his new bicycle. "An old man and a little boy borrowed it," he explained. "they are going to bring it back at four O'clock." His parents were upset that he had given his expensive new bicycle, but were secretly proud of his kindness and faith. Came four O'clock, no bicycle. The parents were anxious. But at 4 : 30, the door bell rang, and there stood a happy man and a boy, with the bicycle and a box of chocolates. Jim suddenly disappeared into his bedroom, and then came running out. "All right," he said. after examining the bicycle. "You can have your watch back!"

73. When Jim came home without his bicycle, his parents -

(a) were angry with him
(b) were worried
(c) did not feel concerned
(d) were eager to meet the old man and the little boy

Ans. (b) : According to the passage, when Jim came home without his bicycle which was given by his parents, they became more worried because that was expensive new bicycle.

74. Jim returned the watch to the old man and the little boy because

(a) they gave him chocolates
(b) his father was proud of him
(c) he was satisfied with the condition of his bicycle
(d) they were late only by 30 minutes

Ans. (c) : According to the passage, at 4: 30 the door bell rang and these stood a happy man and a boy with the bicycle. After examine the bicycle he said "All right" he was satisfied with the condition of his bicycle.

English Passage–2

It was already late when we set out for the next town, which according to the map was about fifteen kilometres away on the other side of the hills. There we felt that we would find a bed for the night. Darkness fell soon after we left the village, but luckily we met no one as we drove swiftly we met no one as we drove swiftly along the narrow winding road that led to the hills. As we climbed higher, it became colder and rain began to fall, making it difficult at times to see the road. I asked John, my companion, to drive more slowly. After we had travelled for about twenty kilometres, there was still no sign of the town which was marked on the map. We were beginning to get worried. Then without warning, the car stopped and we found we had run out of petrol.

75. The author asked John to drive more slowly because -

(a) the road led to the hills
(b) John was an inexperienced driver
(c) the road was not clearly visible
(d) they were in wilderness

Ans. (c) : According to the passage, "we drove swiftly along the narrow winding road that led to the hills. It was colder and rain began to fall, making it difficult at time of see the roads."

76. The travellers set out for the town although it was getting dark because -

(a) they were in a hurry
(b) the next town was a short distance away and was a hill resort
(c) they were in wilderness
(d) the next town was a short distance away and promised a good rest for the night

Ans. (d) : The Travellers set out for the town although it was getting dark because the next town was a short distance away and promised a good rest for the night.

77. The travellers were worried after twenty kilometres because -

(a) it was a lonely countryside
(b) they probably feared of having lost their way
(c) the rain begin to fall
(d) it was getting colder as they drove

Ans. (b) : According to the passage, "when after we had travelled for about twenty kilometers, there was still no sign of the town which was marked on map." Thus, they probably feared of having lost their way.

English Passage–3

A stout old lady was walking with her basket down the middle of a street in Petrograd to the great confusion of the traffic and no small peril to herself. It was pointed out to her that the pavement was the place for foot-passengers, but she replied, "I'm going to walk where I like. We've got liberty now." It did not occur to the dear lady that if liberty entitled the foot-passenger to walk down the middle of the road it also entitled the taxi-driver to drive on the pavement, and that the end of such liberty would be universal chaos. Everything would be getting in everybody else's way and nobody would get anywhere. Individual liberty would have become social anarchy.

78. It was pointed out to the lady that she should walk on the pavement because she was -

(a) a pedestrian
(b) carrying a basket
(c) stout
(d)an old lady

Ans. (a) : It was pointed out to the lady that she should walk on the pavement because she was a pedestrian

79. The lady refused to move from the middle of the street because -

(a) she was not afraid of being killed
(b) she felt that she is entitled to do whatever she liked
(c) she did not like walking on the pavement
(d) she was confused

Ans. (b) : The lady refused to move from the middle of the street because she felt that she is entitled to do whatever she liked.

80. The old lady failed to realise that -

(a) she was not really free
(b) her liberty was not unlimited
(c) she was an old person
(d) roads are made for motor vehicles only

Ans. (b) : According to the passage, "The old lady failed to realize that her liberty was not unlimited."